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Nickel-catalyzed electrochemical carboxylation of propargylic esters with CO_2 to 2,3-allenoic acids

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ABSTRACT

Here, we report a novel nickel-catalyzed electrochemical carboxylation of propargylic esters with CO_2, characterized by the regioselective synthesis of 2,3-allenoic acids rather than propargylic carboxylic acids. Both acyclic propargylic esters and cyclic propargylic carbonates serve as effective substrates, facilitating the synthesis of mono-, di-, tri-, and tetra-substituted 2,3-allenoic acids with broad substrate scope under mild conditions. Mechanistic investigations indicate that the *in situ* generated Ni(I) complex might serve as the active species to react with propargylic esters, forming the allenyl-Ni(I) complex under electro-reductive conditions. A possible γ-selective nucleophilic attack of allenyl-Ni(I) complex on CO_2 is likely involved in the formation of the desired 2,3-allenoic acids.

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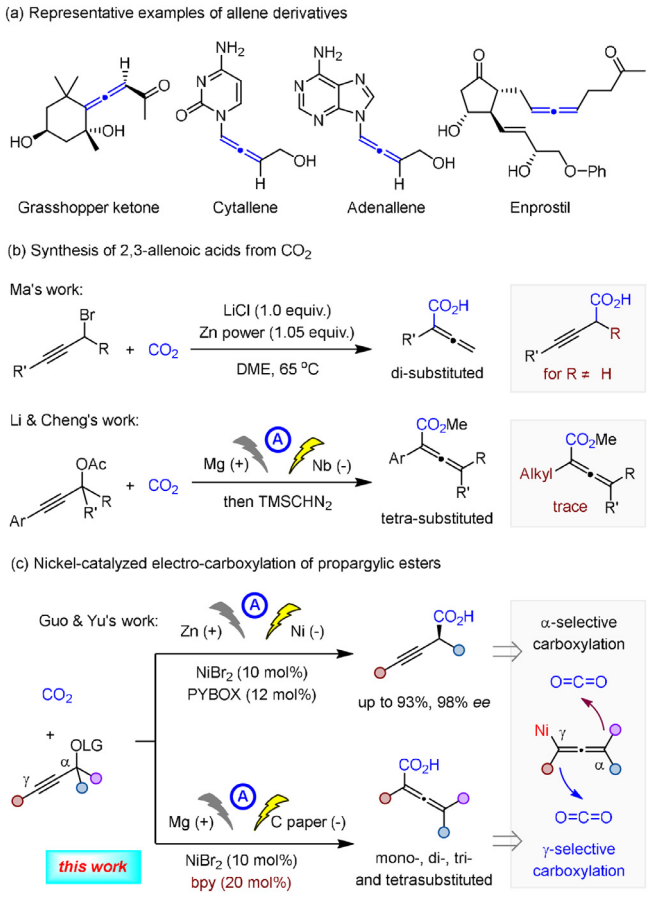
Allenes are common structural motifs in a diverse range of natural products, pharmaceuticals, agrochemicals, and biologically active molecules (Scheme 1a) [1,2]. Their distinctive feature of bearing two cumulated carbon–carbon double bonds make them attractive precursors in various organic transformations [3–8]. The 2,3-allenoic acid, a specific type of functionalized allene that combines the distinctive allenic structure with carboxylic acid functionality, exhibits unique reactivity and biological activities. It can be readily transformed into other allene derivatives, such as allenols, allenals, allenoates, allenamides, and butenolides [9–12]. The effective construction of 2,3-allenoic acids is of great interest, and the conversion of propargylic alcohol derivatives has emerged as a promising synthetic strategy due to the accessibility of these substrates [13–17]. Notable progress has been made in the carboxylation of propargylic alcohols with CO for the construction of 2,3-allenoic acid derivatives [18–28].

Notably, the carboxylation of propargylic alcohol derivatives with CO_2 represents a direct and effective way to construct valuable 2,3-allenoic acids, due to the abundant, nontoxic, and recyclable properties of CO_2 [29–39]. However, the thermodynamic stability and kinetic inertness of CO_2 make it difficult to achieve high efficiency under mild reaction conditions [40–42]. Despite this, seminal research from the Ma [43] and Li & Cheng groups [44] has demonstrated the feasibility of this strategy (Scheme 1b). As early as 2015, Ma developed an elegant Zn-mediated carboxylation of 2-alkynyl bromides with CO_2, enabling the synthesis of 2-substituted 2,3-allenoic acids with high efficiency from primary 2-alkynyl bromides [43]. Notably, the carboxylation of secondary 2-alkynyl bromides yielded 3-alkynoic acids rather than 2,3-allenoic acids, likely due to steric effects. In 2023, Li & Cheng reported the carboxylation of propargylic acetates with CO_2 under an electrochemical reductive system, enabling the construction of tetra-substituted 2,3-allenoic acids under mild conditions [44]. However, likely due to the instability of the radical intermediate, the range of substrates was limited to 3-aryl propargylic acetates, yielding no more than 2-aryl-4,4-dialkyl substituted 2,3-allenoic acids. Despite these advances, both types of reactions are limited to highly reactive substrates and can only yield di- or tetra-substituted 2,3-allenoic acids, respectively. The carboxylation of more challenging materials with CO_2 to construct differently (mono-, di-, tri-, and tetra-) substituted 2,3-allenoic acids with good functional group tolerance remains a highly desirable goal.

With our continuous interest in CO_2 chemical fixation [45–54] and propargylic substitution [55,56], and also encouraged by the advances in electro-reductive [57–71] and metal-catalyzed carboxylation of CO_2 [72–77], we endeavor to develop an electrocatalytic carboxylation of propargylic esters with CO_2, for the diverse construction of differently substituted 2,3-allenoic acids. The combination of electrocatalysis and transition metal catalysis unites

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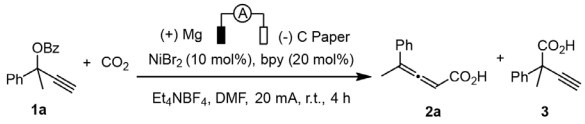


Scheme 1. Carboxylation with CO_2 to 2,3-allenoic acids.

the sustainable nature of electrochemistry with the chemo-, regio-, and stereoselective hallmarks of transition metal-catalyzed synthetic methods, thus providing opportunities for the realization of challenging transformations [78–88]. Recently, Guo & Yu reported an elegant enantioselective Ni-electrocatalyzed reductive propargylic carboxylation of propargylic carbonates with CO_2 (Scheme 1c) [89]. The combination of electro-reduction and Ni-catalysis facilitated the transformation of electrophilic allenyl-Ni(II) into nucleophilic allenyl-Ni(I) species, and the subsequent nucleophilic addition to CO_2 yielded the desired propargylic carboxylic acid with high regio- and enantio-selectivity. On the other hand, for Ni-catalyzed carboxylation with CO_2, the activity and selectivity of the Ni-complex catalyst can be tuned by modifying the electronic and steric properties of the ligands, thus enabling regiodivergent carboxylation [90,91]. In particular, the ligand-controlled regiodivergent Ni-catalyzed carboxylation of alkenes, allyl esters, allylic alcohols and alkyl halide has been realized [92–96]. Accordingly, Ni-catalyzed carboxylation of propargylic esters with CO_2 could not only yield propargylic carboxylic acids but also afford 2,3-allenoic acids *via* regiodivergent manner (Scheme 1c). However, such transformations have never been studied before.

In this context, we envisaged nickel-catalyzed electroreductive carboxylation of propargylic esters with CO_2 would offer a versatile approach to achieve a general and modular catalytic assembly of 2,3-allenoic acids in a regiodivergent manner. Apart from controlling the regioselectivity, avoiding competitive further reductive carboxylation of the allene moiety [97], as well as the application to the synthesis of differently substituted 2,3-allenoic acids, also represent challenges to be overcome to validate this hypothesis. Herein, we present a feasible nickel-catalyzed electro-

Table 1
Condition optimization.



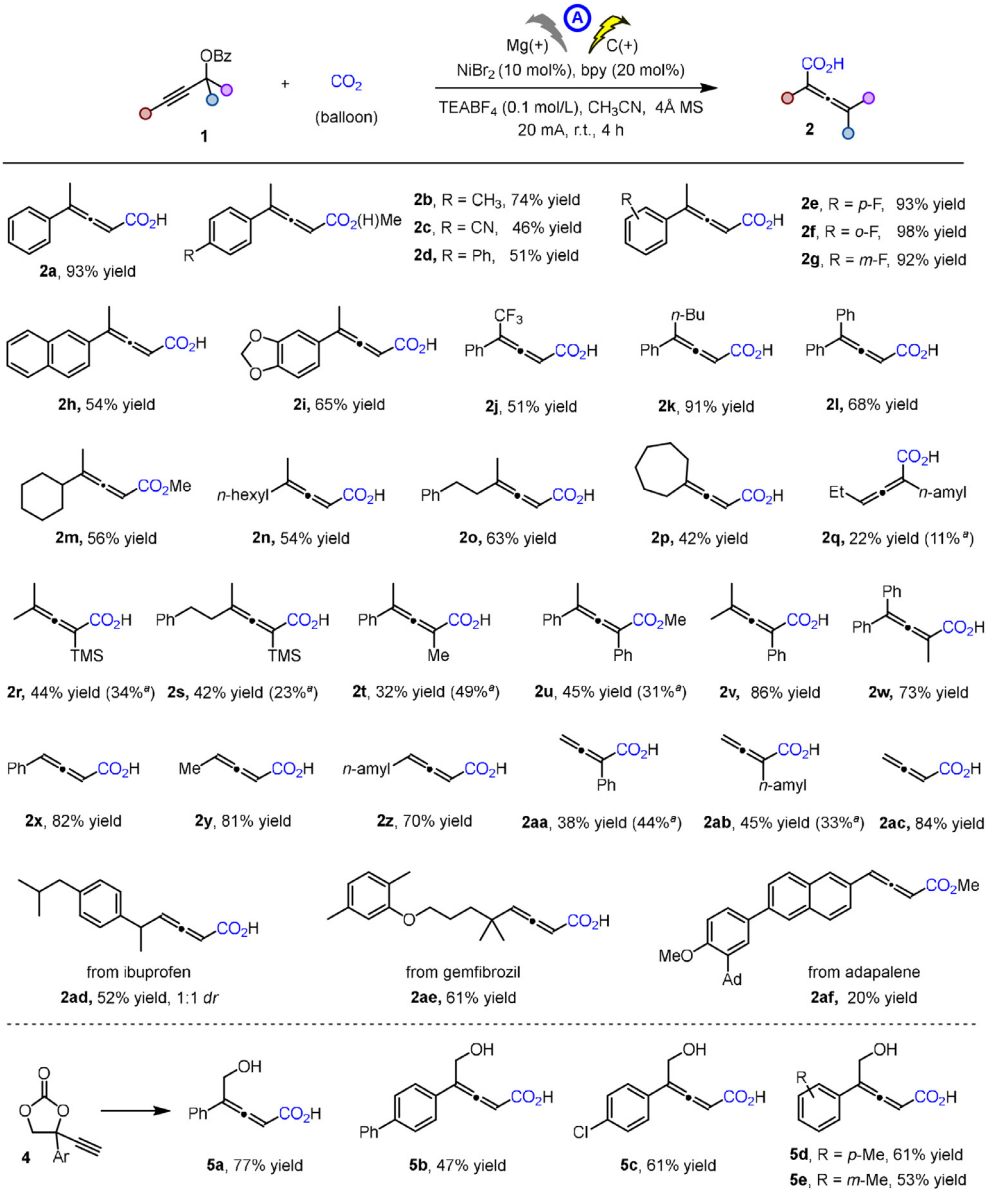
Entry	Variations	Yield (%)[a]	
		2a	**3**
1	None	66	Trace
2	Et_4NClO_4 instead of Et_4NBF_4	58	Trace
3	$^nBu_4NBF_4$ instead of Et_4NBF_4	34	Trace
4[b]	Foam Ni instead of C paper	25	8
5[b]	Al instead of Mg	41	Trace
6[c]	CH_3CN as solvent	93[d]	Trace
7[e]	$Ni(OTf)_2$ instead of $NiBr_2$	86	Trace
8[e]	$NiCl_2$ instead of $NiBr_2$	55	Trace
9[e]	phen instead of bpy	69	Trace
10[e]	tpy instead of bpy	84	Trace
11[e]	10 mol% of bpy	74	Trace

[a] Determined by ^1H NMR with 1,3,5-trimethylbenzene as standard.
[b] Variation of entry 2.
[c] 4 Å MS as additive.
[d] Isolated yield.
[e] Variation of entry 6.

catalytic carboxylation of propargylic esters with CO_2 to produce 2,3-allenoic acids with a broad substrate scope. Not only acyclic propargylic esters but also cyclic propargylic carbonates serve as effective substrates, facilitating the synthesis of mono-, di-, tri-, and tetra-substituted 2,3-allenoic acids effectively under mild conditions (Scheme 1c).

Our study began with the nickel-catalyzed electrochemical carboxylation of propargylic ester **1a** with CO_2 as a model reaction. Following a series of conditional optimizations (for details, see Section 2 in Supporting information), the target tri-substituted 2,3-allenoic acid **2a** was isolated in a 66% yield. The reaction was catalyzed by a $NiBr_2$–bipyridine (bpy) complex under a constant current of 20 mA at room temperature in DMF containing Et_4NBF_4 as the electrolyte, with a C paper as the cathode and a Mg plate as the anode (Table 1, entry 1). The screening of electrolytes by varying the cation or counter anion did not yield improved results (entries 2 and 3). Subsequent studies indicated that the choice of electrode materials significantly influenced the reaction. Using a foam Ni as the cathode or an Al plate as the anode resulted in decreased yields of 25% and 41%, respectively (entries 4 and 5). Fortunately, using CH_3CN as the solvent resulted in a marked increase to 93% yield (entry 6). Under these conditions, further screening of Ni salts with $Ni(OTf)_2$ or $NiCl_2$ as catalysts resulted in decreased yields of 86% and 55%, respectively (entries 7 and 8). The influence of ligands was also studied; changing bpy to 1,10-phenanthroline (phen) or 2,2′:6′,2″-terpyridine (tpy) as ligand resulted in inferior 69% and 84% yields, respectively (entries 9 and 10). The reduction in ligand loading likely impaired the dissolution of Ni salt, thus leading to a decreased 74% yield (entry 11). Ultimately, the conditions shown in entry 6 were selected as the optimized reaction conditions for the nickel-catalyzed electro-carboxylation. Notably, under these conditions, the formation of propargylic carboxylic acid was not detected, as only the 2,3-allenoic acid **2a** was formed, underscoring the exceptional regioselectivity of this Ni-catalyzed electrochemical carboxylation process.

Having established the optimal reaction conditions, we next evaluated the scope of the reaction with respect to propargylic esters (Scheme 2). First, we attempted the synthesis of trisubstituted allenoic acids from propargyl esters. For acetophenone-derived propargyl esters, we found that both electron-withdrawing

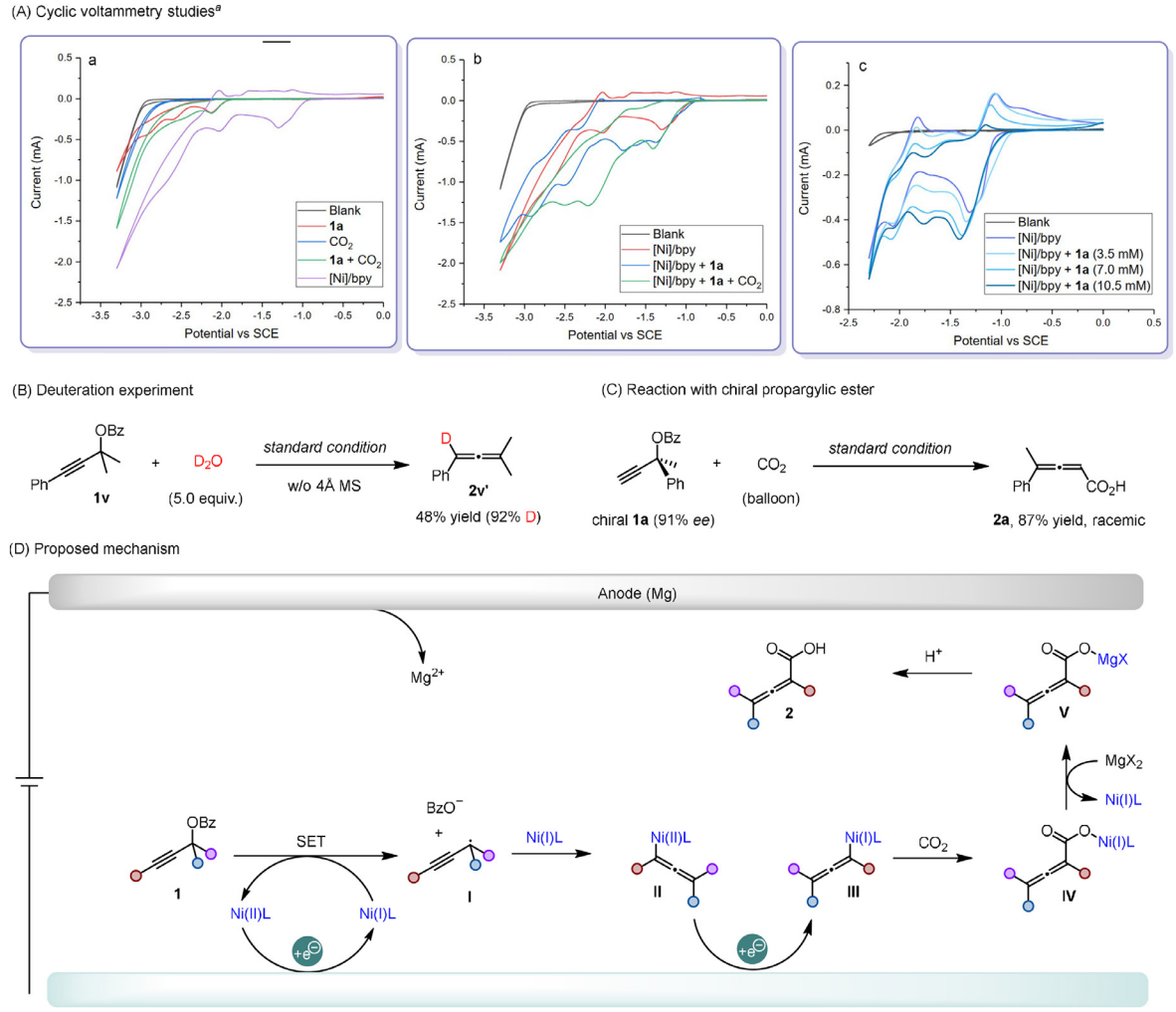


Scheme 2. Substrate scope. Reaction conditions: **1** or **4** (0.2 mmol), Et_4NBF_4 (0.5 mmol), $NiBr_2$ (0.02 mmol), bpy (0.04 mmol), 4 Å MS (200 mg) in 5 mL CH_3CN with CO_2 balloon in undivided cell with Mg anode and C paper cathode, under constant current of 20 mA at room temperature for 4 h. Isolated yields. [a] Yield of the corresponding propargylic carboxylic acids, determined by ^1H NMR analysis of the crude reaction mixture, using 1,3,5-trimethylbenzene as an internal standard.

and electron-donating substituents at the *para*-position on the benzene ring, such as methyl, cyano, phenyl, and fluorine atoms, were suitable for this system, yielding the corresponding 2,4,4-trisubstituted allenoic acids **2a**-**2e** with 46%–93% yield. Similarly, *ortho*- and *meta*-fluoro substituted propargylic esters also worked well, delivering **2f** and **2g** in 98% and 92% yield, respectively. Additionally, various heterocycles, such as naphthyl and piperonyl, were well-tolerated, yielding the corresponding 2,3-allenoic acids **2h** and **2i** in 54% and 65% yield, respectively. Replacing the methyl group with other substituents, such as trifluoromethyl, *n*–butyl, and phenyl, resulted in 2,3-allenoic acids **2j**-**2l** with 51%–91% yield. Later, we tried propargylic esters derived from di-alkyl ketones, which also reacted smoothly, yielding 4,4-dialkyl-substituted 2,3-allenoic acids **2m**-**2p** with 42%–63% yield. The construction of 2,2,4-trisubstituted allenoic acids was also attempted, and starting from *n*-amyl substituted internal propargylic ester **1q**, the desired 2,2,4-trisubstituted allenoic acid **2q** was obtained smoothly, albeit with a promising 22% yield.

After successfully synthesizing trisubstituted 2,3-allenoic acids, we further examined the applicability of this system for the construction of tetrasubstituted 2,3-allenoic acids. Satisfactorily, the reaction of propargylic esters derived from di-alkyl, aryl-alkyl, and di-aryl ketones afforded the corresponding tetrasubstituted 2,3-allenoic acids **2r**-**2w** bearing 2-TMS, 2-methyl, and 2-phenyl substituents with up to 86% yield. This result further demonstrates the broad substrate scope and versatility of this system.

Next, we explored the synthesis of di-substituted 2,3-allenoic acids. The carboxylation of terminal propargylic esters derived from aldehydes with CO_2 proceeded smoothly, yielding the desired 2,4-disubstituted 2,3-allenoic acids **2x**-**2z** bearing 4-phenyl, 4-methyl, and 4-*n*-amyl substituents with 70%–82% yield. Meanwhile, the reaction of internal propargylic esters derived from formaldehyde delivered the 2,2-disubstituted 2,3-allenoic acids **2aa** and **2ab** with 2-phenyl and 2-*n*-amyl substituents in 38% and 45% yield, respectively. Gratifyingly, structurally complex propargylic



Scheme 3. Mechanistic study. [a] The CV was performed in DMF with 0.1 mol/L Et_4NBF_4, 100 mV/s using glassy carbon as working electrode.

esters derived from drugs such as ibuprofen, gemfibrozil (Lopid), and adapalene also afforded the corresponding 2,4-disubstituted 2,3-allenoic acids **2ad**-**2af** with good yields. Lastly, the construction of the simplest monosubstituted 2,3-allenoic acid was attempted, and the reaction of 2-propynyl benzoate successfully afforded 2,3-butadienoic acid **2ac** in 84% yield.

Motivated by the successful electrochemical carboxylation of propargylic esters, we extended our investigation to cyclic propargylic carbonates, aiming to synthesize methylol-substituted 2,3-allenic acids. To our delight, the reaction with α-ethynyl cyclic carbonate was successful, affording a series of trisubstituted 2,3-allenic acids **5a**-**5e** featuring a 4-methylol moiety with 47%–77% yield. Through the evaluation of various propargyl alcohol derivatives, we believe that the reaction under electrochemical conditions exhibits broad substrate versatility and application prospects, adding a new method for synthesizing a series of structurally diverse 2,3-allenoic acids.

It should be pointed out that during the synthesis of typical 2-trisubstituted 2,3-allenoic acids, such as **2q**-**2u** and **2aa**-**2ab**, the generation of corresponding propargylic acids could be detected. This is likely due to the steric hindrance effect of the γ-selective carboxylation of allenyl-Ni species, which presents a limitation of this method (For detail, please see Section 5 in Supporting information).

To elucidate the reaction mechanism, cyclic voltammetry (CV) analyses were initially conducted to study the electrochemical process at the cathode. The results revealed that the nickel complex exhibits two reduction potential peaks: The first at −1.3 V and the second at −2.0 V, corresponding to the Ni(II)/Ni(I) and Ni(I)/Ni(0) processes, respectively. The first reduction potential peak is more positive than that of CO_2 and propargylic ester **1a**, indicating that during the reaction process, the Ni(II) complex is initially reduced to Ni(I) (Scheme 3A-a). Upon the addition of **1a** to the nickel complex, the current at −1.3 V increases significantly, indicating that the Ni(I) complex generated after the first (single-electron transfer) SET reduction should interact with **1a** immediately, leading to an increase in the current. During this process, new complexes might be formed and subsequently further electrolyzed, giving rise to new reduction peaks. Further introduction of CO_2 shows no obvious changes in the reduction peak at −1.3 V, while the current at the secondary reduction peak increases significantly, indicating that the new species generated after the secondary reduction should react with CO_2 immediately (Scheme 3A-b). Furthermore, the influence of the concentration of **1a** on the reduction of the Ni complex was studied. As the concentration of **1a** increases, the current at −1.3 V increases, whereas there is no substantial increase in current at −2.0 V (Scheme 3A-c). This further substantiates that it shoulbe the Ni(I) species, rather than the Ni(0) species, that reacts with propargylic ester **1a** during the reaction course.

To gain more insights into the reaction process, the following control experiments were conducted. First, a deuteration experiment with compound **1v** was performed under optimal conditions

using 5.0 equiv. of D_2O (Scheme 3B). The experiment resulted in the detection of allene **2v'** in 48% yield with a 92% deuteration ratio, implying the possible involvement of allenyl-Ni(I) intermediates during the transformation. Furthermore, the carboxylation of chiral propargylic ester **1a** under standard conditions yielded the racemic product **2a** in 87% yield (Scheme 3C). This outcome suggests that the C–O bond cleavage is chirality-unretained, and a radical process might be involved.

Based on experimental outcomes and literature reports on the Ni-catalyzed coupling reaction of sp^3 electrophiles [98–102], particularly those involving C(sp^3)–O bond transformations [103–105], a plausible mechanism for the Ni-catalyzed electrochemical carboxylation of propargylic esters with CO_2 to form 2,3-allenoic acids is proposed (Scheme 3D). Initially, the catalyst precursor, a Ni(II) complex, is reduced at the cathode to afford a Ni(I) species. This Ni(I) species reacts with propargylic esters **1** to generate a radical intermediate **I** and regenerate the Ni(II) species, which could be further reduced to Ni(I) species at the cathode. The interaction of Ni(I) species with radical intermediate **I** resulting the generation of allenyl-Ni(II) species **II**, which undergoes a further SET reduction to form the nucleophilic allenyl-Ni(I) complex **III**. Finally, nucleophilic attack at the γ-position of **III** with CO_2 produces an intermediate **IV**, which upon acidification yields the 2,3-allenoic acid product **2**. At the anode, magnesium oxidizes to Mg^{2+}, which could stabilize carboxylate intermediates through electrostatic interactions and may also enhance reaction efficiency by increasing electrolyte conductivity. Additionally, the magnesium electrode offers a high reducing potential, making it superior to other metal anodes.

In conclusion, we have successfully developed an electrochemical approach for the Ni-catalyzed carboxylation of propargylic esters with CO_2, achieving the selective construction of 2,3-allenoic acids. This process offers high regioselectivity and broad substrate tolerance, with both propargylic esters and cyclic carbonates serving as viable substrates. It enables the synthesis of mono-, di-, tri-, and tetra-substituted 2,3-allenoic acids with diverse structures. Mechanistic investigations support a pathway involving the generation of a Ni(I) complex as the active species for the reaction with propargylic esters and a possible γ-selective nucleophilic attack on CO_2 of the allenyl-Ni(I) complex. The extension of this electrocatalytic carboxylation strategy for the synthesis of various carboxylic acids through reductive C–O bond cleavage is currently underway.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yuqing Zhong: Writing – original draft, Methodology, Investigation, Data curation. **Mengmeng Jiang:** Investigation, Data curation. **Deyong Yang:** Investigation, Data curation. **Nan Feng:** Investigation, Data curation. **Ying Sun:** Investigation, Data curation. **Huimin Wang:** Investigation, Data curation. **Feng Zhou:** Writing – review & editing, Project administration, Methodology.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cclet.2025.111169.

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